

volatility is comprised of both market risk and company specific risk. The ETF price sensitivity a_1 parameter needs to be higher to avoid a potential arbitrage opportunity. For example, suppose volumes are identical across stocks and ETF (we are ignoring shadow liquidity here for simplicity). If the manager invests the same dollar amount in the ETF or in the underlying stock portfolio (following the same weightings) the trading costs should be the same. But since the ETF volatility will be less than the weighted stock portfolio the market impact model will estimate lower costs for the ETF than the portfolio of stocks. To correct for the mispricing, a proper ETF market impact model will need to have a higher sensitivity term as follows:

$$a_{1ETF} = a_{1Stock} \cdot \left(\frac{w'\sigma}{\sigma_p} \right)^{a_3}$$

where $w'\sigma$ is the weighted volatility of the ETF portfolio and σ_p is the ETF portfolio volatility incorporating all covariance and correlation benefits.

Futures: The initial cost of trading futures is usually the highest compared to ETFs and equities. The reason is primarily due to the roll cost associated with purchasing the next futures contract at the time of the contract expiration. Managers who maintain a portfolio of futures contracts will need to continuously purchase the next contract. This creates a recurring cost. An advantage of trading futures is that the contract sizes are usually extremely large in size and trading costs increase at the slowest rate compared to ETFs and equities. Investors will transact in futures for various reason. First investors purchase futures to hedge positions. Second, managers purchase futures contracts often for speculation. Third, investors have begun investing in futures portfolios due to the cost advantage they provide for very large orders. All of these investment reasons provide a great deal of liquidity for futures contracts and result in lower incremental trading costs.

Beta Investment Allocation

A common misconception with beta allocation investment strategies is that the portfolio manager should purchase only a single investment vehicle to achieve their exposure. For example in [Figure 12.7a](#) we found that equities are the most cost efficient vehicle if dollar value is less than \$250 million. ETFs are the most cost effective vehicle if dollar value is between \$250 and \$667 million, and futures are the most cost effective if dollar value is greater than \$667 million. But this is only true if the